

Part 2: Single-Variable Integral Calculus

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- Given the velocity of a function, one wishes to know its position at a given time.
- Find a function whose derivative is known.

Definition 1

A function $F(x)$ is called an *antiderivative of $f(x)$ on an interval I* if $F'(x) = f(x)$ for all $x \in I$.

Theorem 1.1

If $F(x)$ is an antiderivative of $f(x)$ on I . Then the family of all antiderivatives of $f(x)$ is $F(x) + C$.

Definition 2

The family of all antiderivatives of is called *the indefinite integral* of $f(x)$. Given an antiderivative $F(x)$ of $f(x)$:

$$\int f(x)dx = F(x) + C.$$

Example 1

$\ln(1 + x^2)$ is an antiderivative of $\frac{2x}{1 + x^2}$, hence

$$\int \frac{2x}{1 + x^2} dx = \ln(1 + x^2) + C.$$

Theorem 1.2

A function $f(x)$ which is continuous on $[a, b]$ has an antiderivative on that interval.

Theorem 1.3

Let $F(x)$, $G(x)$ be antiderivatives of $f(x)$, $g(x)$ respectively. Then

$$\int (Af(x) + Bg(x))dx = AF(x) + BG(x) + C.$$

Table of indefinite integrals

$$\int x^{\alpha} dx = \begin{cases} \frac{x^{\alpha+1}}{\alpha+1} + C & \text{if } \alpha \neq -1, \\ \ln |x| + C & \text{if } \alpha = -1. \end{cases}$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, 0 < a \neq 1$$

$$\int \sin x dx = -\cos x + C, \quad \int \cos x dx = \sin x + C$$

$$\int \frac{dx}{\sin^2 x} = -\cot x + C, \quad \int \frac{dx}{\cos^2 x} = \tan x + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C$$

Example 2

The linear density of a rod of length 4m is given by $\rho(x) = 9 + 2\sqrt{x}$ measured in kilograms per meter, where x is measured in meters from one end of the rod. Find the total mass of the rod.

The mass of the rod which is x meters long is

$$m = \int (9 + 2\sqrt{x})dx = 9x + \frac{4}{3}x\sqrt{x} + C.$$

$$x = 0 \Rightarrow C = 0.$$

$$x = 4 \Rightarrow m = 9 \cdot 4 + \frac{4}{3} \cdot 4\sqrt{4} \approx 46,6(\text{kg})$$

Theorem 1.4

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

Theorem 1.5

If f is continuous on an interval I , $x(t) : J \rightarrow I$ has an inverse and $x'(t)$ is continuous on J , then

$$\int f(x)dx = \int f(x(t))x'(t)dt.$$

If $\int f(x(t))x'(t)dt = G(t) + C$, then $\int f(x)dx = G(t(x)) + C$.

Example 3

Evaluate the following integrals

$$1 \quad \int \frac{dx}{4x^2 + 9}$$

$$2 \quad \int \frac{dx}{\sqrt{2x - x^2}}$$

$$3 \quad \int \frac{dx}{x \ln x}$$

$$4 \quad \int \frac{x^4 dx}{x^{10} + 1}$$

$$5 \quad \int \frac{e^x dx}{\sqrt{4 - e^{2x}}}$$

$$6 \quad \int \frac{dx}{\sqrt{x^2 + 4}}$$

Integration by parts

Assume that $u(x)$, $v(x)$ are continuously differentiable functions. We have

$$\int u dv = uv - \int v du.$$

Example 4

Evaluate the integrals

7 $\int x \arctan x dx$

8 $\int x \ln(1 + 2x) dx$

9 $\int \sqrt{x^2 + 4} dx$

Example 5

Evaluate the integrals

$$10 \quad \int \frac{(3x + 1)dx}{2x^2 - 5x + 2}$$

$$11 \quad \int \frac{x dx}{(x + 2)^2(x - 1)}$$

$$12 \quad \int \frac{(2x + 1)dx}{x^2 + 9}$$

$$13 \quad \int \frac{(x + 5)dx}{x^3 - 8}$$

Aim: Evaluate $\int R(x)dx$, where

$$R(x) = \frac{b_0 + b_1x + \dots + b_mx^m}{a_0 + a_1x + \dots + a_nx^n}, a_i, b_i \in \mathbb{R}, a_n, b_m \neq 0,$$

is a rational function.

Method: expressing $R(x)$ as a sum of partial fractions.

- ❶ Performing a polynomial division to reduce to a *proper* rational function ($m < n$).
- ❷ Factorizing the denominator into factors $(x - a)^k, (x^2 + px + q)^k$, where $q - \frac{p^2}{4} > 0$.
- ❸ Writing $R(x)$ as a sum of partial fractions.

$$\int \frac{A_l dx}{(x - a)^l}, \quad \int \frac{B_l x + C_l}{(x^2 + px + q)^l} dx, l = 1, 2, \dots, k.$$

Trigonometric integrals $\int \mathcal{R}(\sin x, \cos x) dx$

Evaluate $\int \mathcal{R}(\sin x, \cos x) dx$, where $\mathcal{R}(\sin x, \cos x)$ is a rational function of the variables $u = \sin x$, $v = \cos x$.

General substitution: $t = \tan \frac{x}{2}$, $x \in (-\pi, \pi)$.

We obtain $\int \mathcal{R}\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2dt}{1+t^2}$.

Example 6

Evaluate the following integrals

14 $\int \frac{dx}{3 \sin x + 4 \cos x}.$

15 $\int \frac{dx}{2 \sin x - \cos x + 5}.$

- If $\mathcal{R}(-\sin x, -\cos x) = \mathcal{R}(\sin x, \cos x)$, **set** $t = \tan x$ **or** $t = \cot x$.
- If $\mathcal{R}(-\sin x, \cos x) = -\mathcal{R}(\sin x, \cos x)$, **set** $t = \cos x$.
- If $\mathcal{R}(\sin x, -\cos x) = -\mathcal{R}(\sin x, \cos x)$, **set** $t = \sin x$.

Example 7

Evaluate the following integrals

16 $\int \sin^2 x \cdot \cos^5 x dx$

17 $\int \frac{dx}{\sin x}$

18 $\int \frac{\sin^3 x}{\cos^5 x} dx$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C$$

$$\int \frac{dx}{\sqrt{x^2 + \alpha}} = \ln |x + \sqrt{x^2 + \alpha}| + C$$

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a} + C$$

$$\int \sqrt{x^2 + \alpha} dx = \frac{x}{2} \sqrt{x^2 + \alpha} + \frac{\alpha}{2} \ln |x + \sqrt{x^2 + \alpha}| + C.$$

Consider $\int \mathcal{R}(x, \sqrt{a^2 - x^2})dx$, $\int \mathcal{R}(x, \sqrt{x^2 + a^2})dx$.

Expression	Substitution	Result
$\sqrt{a^2 - x^2}$	$x = a \sin t, t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$a \cos t$
$\sqrt{a^2 + x^2}$	$x = a \tan t, t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$\frac{a}{\cos t}$
$\sqrt{x^2 - a^2}$	$x = \frac{a}{\cos t}, t \in \left[0, \frac{\pi}{2}\right)$	$a \tan t$

Example 8

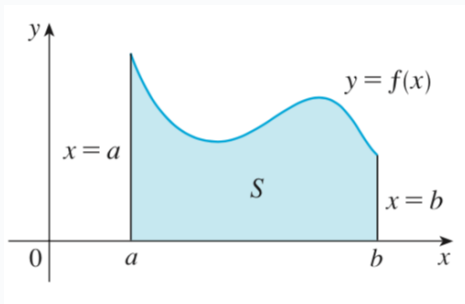
Evaluate the following integrals

19 $\int \frac{\sqrt{9 - x^2}}{x^2} dx$

20 $\int \frac{\sqrt{x^2 - 1}}{x^4} dx$

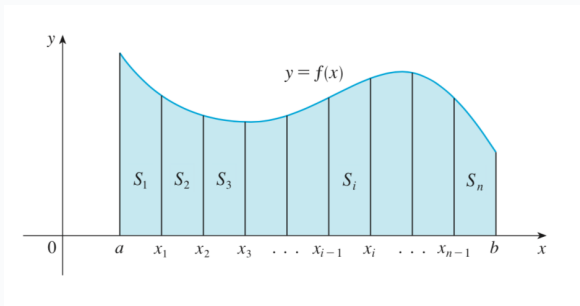
21 $\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$

The area problem



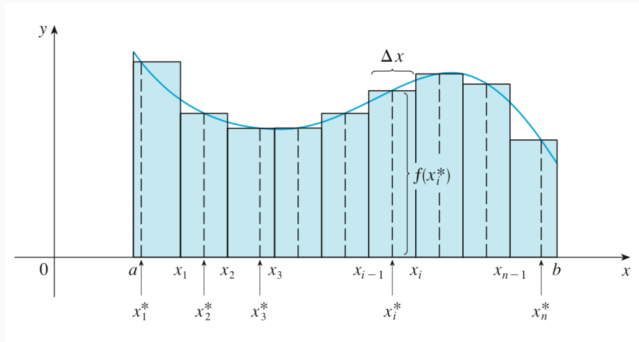
Hình: $S = \{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x)\}.$

Problem: Find the area of the region S under the curve $y = f(x)$ from a to b , ($f(x) \geq 0$).



A partition of $[a, b]$: $a \equiv x_0 < x_1 < \dots < x_n \equiv b$, $\Delta x_i = x_i - x_{i-1}$.

$$S = S_1 + S_2 + \dots + S_n = \sum_{k=1}^n S_k.$$



Pick a sample point $x_k^* \in [x_{k-1}, x_k]$, $1 \leq k \leq n$; $S_k \approx f(x_k^*)\Delta x_k$.

$$S \approx \sum_{k=1}^n f(x_k^*)\Delta x_k.$$

$$S := \lim_{\substack{n \rightarrow \infty \\ \lambda \rightarrow 0}} \sum_{k=1}^n f(x_k^*)\Delta x_k \text{ if the limit exists, } \lambda = \max_{1 \leq k \leq n} \Delta x_k.$$

The distance problem

Problem: Find the distance traveled by an object during a time period $a \leq t \leq b$, given the velocity $v = v(t)$.
Easy situations

1 $v(t) = v_0 = \text{constant} \Rightarrow s = v_0(b - a)$.

2

Time (s)	0	5	10	15	20	25
Velocity (m/s)	10	12	15	18	14	14

The distance traveled is

$$s \approx (10 \times 5) + (12 \times 5) + (15 \times 5) + (18 \times 5) + (14 \times 5) + (14 \times 5).$$

Divide $[a, b]$ into subintervals $a \equiv t_0 < t_1 < \dots < t_n \equiv b$, $\Delta t_k = t_k - t_{k-1}$.
 Pick a sample time $t_k^* \in [t_{k-1}, t_k]$.

The approximate distance is $\sum_{k=1}^n v(t_k^*) \cdot \Delta t_k$.

The distance traveled is

$$s = \lim_{\substack{n \rightarrow \infty \\ \lambda \rightarrow 0}} \sum_{k=1}^n v(t_k^*) \Delta t_k, \quad \lambda = \max_{1 \leq k \leq n} \Delta t_k,$$

if the limit exists.

Definition 3

If $\lim_{\substack{n \rightarrow \infty \\ \lambda \rightarrow 0}} \sum_{k=1}^n f(x_k^*) \Delta x_k = I$ exists, then I is called the **definite integral** of $f(x)$ on $[a, b]$.

Notation $I = \int_a^b f(x) dx$.

$f(x)$: **integrand**, a : **lower limit**, b : **upper limit**.

$f(x)$ is said to be **integrable** on $[a, b]$.

The area of the region under the graph of $f(x) \geq 0$ from a to b is $S = \int_a^b f(x) dx$.

Theorem 2.1

If $f(x)$ is integrable on $[a, b]$, then $f(x)$ is bounded.

Theorem 2.2

If $f(x)$ is continuous on $[a, b]$, then it is integrable on that interval.

Theorem 2.3

If $f(x)$ is bounded on $[a, b]$, and has finitely many discontinuities then it is integrable on that interval.

Example 9

$f(x) = \begin{cases} -1, & \text{if } -1 \leq x \leq 0, \\ 1, & \text{if } 0 < x \leq 1, \end{cases}$ is integrable on $[-1, 1]$.

Let $f(x), g(x)$ be two integrable functions on $[a, b]$.

$$\textcircled{1} \quad \int_b^a f(x)dx = - \int_a^b f(x)dx.$$

$\textcircled{2}$ Linearity: for constants A, B

$$\int_a^b (Af(x) + Bg(x))dx = A \int_a^b f(x)dx + B \int_a^b g(x)dx.$$

$\textcircled{3}$ Let $c \in (a, b)$,

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

$\textcircled{4}$ If $f(x) \geq 0$ for $x \in [a, b]$, then

$$\int_a^b f(x)dx \geq 0.$$

Corollary 2.1

If $f(x) \leq g(x)$ for $x \in [a, b]$, then

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx.$$

In particular,

$$\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx.$$

Without evaluating the integrals, we know

$$0 \leq \int_0^{\frac{\pi}{2}} \sin^4 x dx \leq \int_0^{\frac{\pi}{2}} \sin^3 x dx \leq 1.$$

Theorem 2.4 (Mean value theorem)

If $m \leq f(x) \leq M$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a).$$

There exists $\mu \in [m, M]$ such that

$$\int_a^b f(x)dx = \mu(b-a).$$

Moreover, if $f(x)$ is continuous on $[a, b]$ then there exists $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x)dx.$$

Theorem 2.5

If $f(x)$ is continuous on $[a, b]$, then

$$F(x) = \int_a^x f(t)dt, a \leq x \leq b,$$

is well-defined on $[a, b]$. Moreover,

$$\left(\int_a^x f(t)dt \right)' = F'(x) = f(x), \forall x \in (a, b).$$

Corollary 2.2

Let $\alpha(x), \beta(x), f(x)$ be continuous on $[a, b]$, and $\alpha(x), \beta(x)$ are differentiable functions. Then

$$\frac{d}{dx} \left(\int_{\alpha(x)}^{\beta(x)} f(t)dt \right) = f(\beta(x))\beta'(x) - f(\alpha(x))\alpha'(x).$$

Example 10

a) Find the derivative of the given function.

1 $F(x) = \int_x^1 t^2 \sin t dt.$

2 $G(x) = \int_0^{x^4} \arctan t dt.$

b) Compute the limit $\lim_{x \rightarrow 0} \frac{\int_0^{\sin x} t^2 \sin t dt}{x^4}$

Theorem 2.6 (Newton-Leibniz formula)

If $f(x)$ is continuous on $[a, b]$ and $F(x)$ is an antiderivative of $f(x)$, then

$$\int_a^b f(x)dx = F(x) \Big|_a^b = F(b) - F(a).$$

Example 11

Express the following limits in terms of definite integrals

$$\textcircled{1} \quad L_1 = \lim_{n \rightarrow \infty} \frac{1}{n} \left(1 + \sqrt{1 + \frac{1}{n}} + \dots + \sqrt{1 + \frac{n-1}{n}} \right)$$

$$\textcircled{2} \quad L_2 = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n \frac{k^3}{n + 2k}.$$

Theorem 2.7

If $u = g(x): [a, b] \rightarrow [g(a), g(b)]$ is continuously differentiable on $[a, b]$, f is continuous on $[g(a), g(b)]$, then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Theorem 2.8

If $x = x(t): [\alpha, \beta] \rightarrow [a, b]$ is continuously differentiable on $[a, b]$, $x(\alpha) = a$, $x(\beta) = b$, f is continuous on $[a, b]$, then

$$\int_a^b f(x)dx = \int_{\alpha}^{\beta} f(x(t))x'(t)dt.$$

Example 12

Evaluate the integrals

$$1 \quad \int_0^1 e^x \cos^5(e^x) dx$$

$$2 \quad \int_1^e \frac{dx}{x(\ln^2 x + 3 \ln x + 2)}$$

$$3 \quad \int_0^{2/\sqrt{3}} \frac{x^2}{(x^2 + 4)^2} dx$$

$$4 \quad \int_0^{\pi} \sin^3 x \cos^2 x dx.$$

Example 13

- ❶ If $f(x)$ is an even function on $[-l, l]$, then

$$\int_{-l}^l f(x)dx = 2 \int_0^l f(x)dx.$$

- ❷ If $f(x)$ is an odd function on $[-l, l]$, then

$$\int_{-l}^l f(x)dx = 0.$$

Theorem 2.9

Let $u(x), v(x)$ be continuously differentiable on $[a, b]$. We have

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du.$$

Example 14

Evaluate the integrals

1 $\int_0^1 x \arctan x dx$

2 $\int_0^1 \arccos^2 x dx.$

Area enclosed by curves

- The area enclosed by $y = f(x)$, $y = g(x)$ and $x = a$, $x = b$ is:

$$S = \int_a^b |f(x) - g(x)| dx.$$

- The area enclosed by $x = f(y)$, $x = g(y)$ and $y = c$, $y = d$ is:

$$S = \int_c^d |f(y) - g(y)| dy.$$

Example 15

- The area enclosed the curves $y = x - 1$, $y^2 = 2x + 6$.

Points of intersection $(-1, -2)$; $(5, 4)$. $V = \int_{-2}^4 \left| (y + 1) - \frac{y^2 - 6}{2} \right| dy.$

- The area enclosed by the curve $y = \sin x$ and the lines $y = 0$, $x = 0$, $x = \frac{3\pi}{2}$.

$$V = \int_0^{\frac{3\pi}{2}} |\sin x| dx.$$

Recall: Parametric curves

Assume that $f(t), g(t)$ are functions of the third variable, the **parameter** t . For each t , we determine a point $M(f(t), g(t))$. When t varies, M also varies and traces out a **parametric curve** C .

Parametrization of (C) : $\begin{cases} x = f(t) \\ y = g(t). \end{cases}$

Example 16

❶ *The unit circle* $\begin{cases} x = \cos t \\ y = \sin t, 0 \leq t \leq 2\pi. \end{cases}$

❷ *The parametric equation of a line:* $\begin{cases} x = 2t + 1 \\ y = -3t - 1, t \in \mathbb{R}. \end{cases}$

If $f'(t) \neq 0$, then $\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$.

Areas enclosed by Parametric curves

- The area between the curve $x = x(t), y = y(t), (\alpha \leq t \leq \beta)$, (which traverses once as $t \in [\alpha, \beta]$) and Ox is

$$S = \int_{\alpha}^{\beta} |y(t)x'(t)| dt.$$

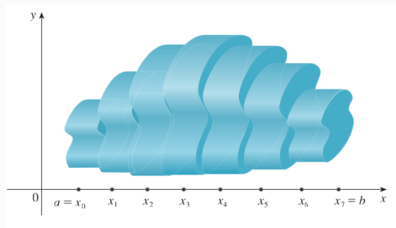
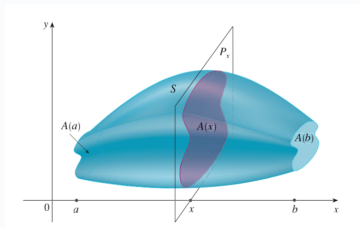
- The area enclosed by the rays $\theta = \alpha, \theta = \beta, (\beta - \alpha \leq 2\pi)$ and the polar curve $r = r(\theta)$ is:

$$S = \frac{1}{2} \int_{\alpha}^{\beta} r^2(\theta) d\theta.$$

Example 17

Find the areas enclosed by

- the astroid $x = 8 \cos^3 t, y = 8 \sin^3 t$.
- the cardioid $r = 1 + \cos \theta$.



Consider a solid S that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane (P_x), through x and perpendicular to the x -axis is $A(x)$.

Divide S into n slabs of width Δx_i , we can approximate the volume S_i by a cylinder with base area $A(x_i^*)$ and

height Δx_i : $V \approx \sum_{i=1}^n A(x_i^*) \Delta x_i \rightarrow \int_a^b A(x) dx$.

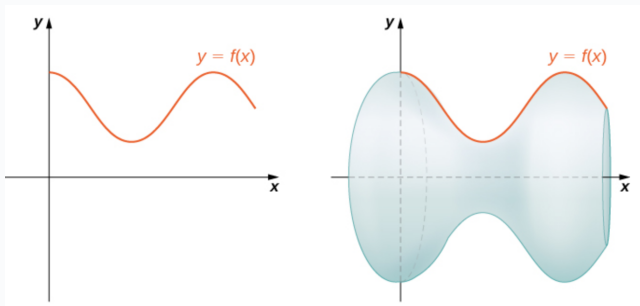
Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane (P_x) , through x and perpendicular to the x -axis is $A(x)$, then the volume of S is

$$V = \int_a^b A(x) dx.$$

Solids of revolution

Solids of revolution are obtained by revolving a region about a line.

Rotating the graph $y = f(x)$, $a \leq x \leq b$ about Ox



The cross section is a circle of radius $|f(x)|$, so $A(x) = \pi f^2(x)$.

The volume of the solid obtained by rotating the region bounded by $x = a$, $x = b$, $y = 0$ and $y = f(x)$ about Ox is

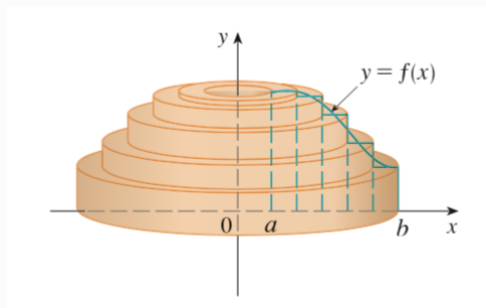
$$V = \pi \int_a^b f^2(x) dx.$$

The volume of the solid obtained by rotating the region bounded by $y = c$, $y = d$, $x = 0$ and $x = g(y)$ about Oy is

$$V = \pi \int_c^d g^2(y) dy.$$

Given a function $f(x) > 0$ on $[a, b]$ where $0 < a < b$. The volume of the solid obtained by rotating the region bounded by $y = f(x)$, $y = 0$, $x = a$, $x = b$ about Oy is

$$V = 2\pi \int_a^b x f(x) dx.$$



Example 18

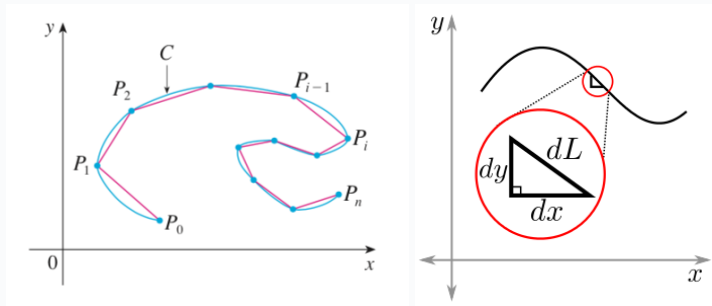
Find the volume of the solids of revolution obtained by rotating

- ❶ *the region enclosed by $y = \sin x$ and the lines $y = 0, x = 0, x = \frac{\pi}{2}$ about Ox .*
- ❷ *the region enclosed by $y = x^2$ and the lines $y = 0, y = 1, x = 0$ about Oy .*
- ❸ *the region enclosed by $y = x^2$ and the lines $y = 0, x = 0, x = 1$ about Oy .*

$$\text{❷ } V_2 = \pi \int_0^1 (\sqrt{y})^2 dy.$$

$$\text{❸ } V_3 = 2\pi \int_0^1 x \cdot x^2 dx.$$

Arclength



$$L = \sum_{i=1}^n \widehat{P_{i-1}P_i} \approx \sum_{i=1}^n P_{i-1}P_i = \sum_{i=1}^n \sqrt{1 + [f'(x_i^*)]^2} \Delta x_i \rightarrow \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

The arclength of the curve

- $(C) : y = f(x), a \leq x \leq b,$

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

- $(C) : \begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad (\alpha \leq t \leq \beta)$

$$L = \int_{\alpha}^{\beta} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt.$$

- $(C) : r = r(\theta), \alpha \leq \theta \leq \beta,$

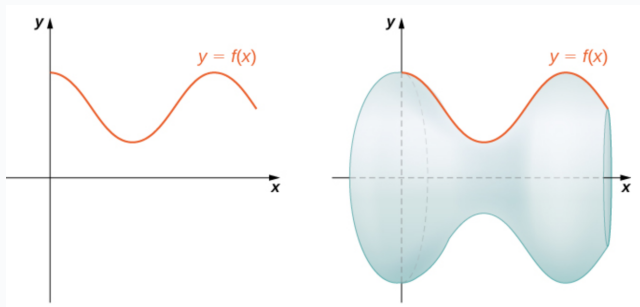
$$L = \int_{\alpha}^{\beta} \sqrt{r^2(\theta) + [r'(\theta)]^2} d\theta.$$

Example 19

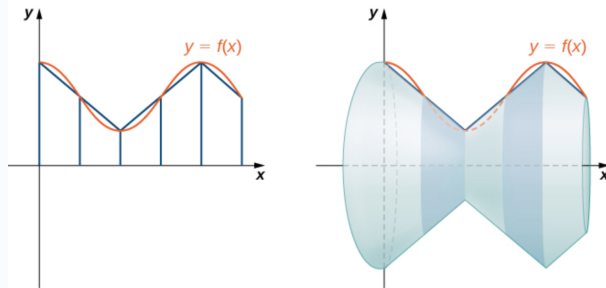
- ① Find the arclength of the curve $y = x^{\frac{3}{2}}$, $0 \leq x \leq 4$.
- ② Find the arclength of the curve $x = a \cos^3 t$, $y = a \sin^3 t$, $a > 0$.
- ③ Find the arclength of the curve $r = a(1 + \cos \theta)$, ($a > 0$).

Lateral surface area of solids of revolution

Rotating the graph $y = f(x)$, $a \leq x \leq b$ about Ox



Hình: Source: <https://math.libretexts.org/>



Hình: Source: <https://math.libretexts.org/>

The lateral surface area of the frustum of a circular cone with upper radius r_1 and lower radius r_2 and lateral height l is $A = 2\pi r l, r = \frac{r_1 + r_2}{2}$.

The lateral surface area of the solid of revolution obtained by

- rotating the graph $y = f(x)$, $a \leq x \leq b$ about Ox is

$$S = 2\pi \int_a^b |f(x)| \sqrt{1 + f'^2(x)} dx.$$

- rotating the curve $x = g(y)$, $c \leq y \leq d$ about Oy is

$$S = 2\pi \int_c^d |g(y)| \sqrt{1 + g'^2(y)} dy.$$

Example 20

Find the surface area of the solid of revolution obtained by rotating the graph $y = \sin x$, $0 \leq x \leq \frac{\pi}{2}$ about Ox .

Improper integrals of type 1

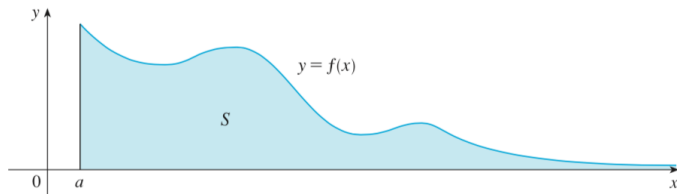
Definition 4

Let $f(x)$ be defined on $[a, +\infty)$ and **integrable** on the interval $[a, A]$, $a \leq A < \infty$.

The **improper integral** of $f(x)$ on $[a, +\infty)$:

$$\int_a^{+\infty} f(x)dx = \lim_{A \rightarrow +\infty} \int_a^A f(x)dx.$$

If the limit exists (as a finite number), we say the integral **converges**. Otherwise, we say it **diverges**.



Hình: Geometric interpretation

Example 21

Evaluate the improper integrals

$$1 \quad \int_0^{+\infty} \sin x dx$$

$$2 \quad \int_1^{+\infty} \frac{dx}{x^\alpha}.$$

$$\int_1^{+\infty} \frac{dx}{x^\alpha} = \lim_{A \rightarrow +\infty} \int_1^A \frac{dx}{x^\alpha}. \quad I_A = \int_1^A \frac{dx}{x^\alpha} = \begin{cases} \ln A & \text{if } \alpha = 1 \\ \frac{A^{1-\alpha}-1}{1-\alpha} & \text{if } \alpha \neq 1. \end{cases}$$

The limit exists iff $\alpha > 1$. Then, $I = \frac{1}{\alpha-1}$.

Remark 3.1

$$\int_a^{+\infty} \frac{dx}{x^\alpha}, (a > 0), \text{ converges} \Leftrightarrow \alpha > 1.$$

Definition 5

① If $\int_{A'}^a f(x)dx$ exists for all $A' \leq a$, then:

$$\int_{-\infty}^a f(x)dx = \lim_{A' \rightarrow -\infty} \int_{A'}^a f(x)dx$$

if the limit exists as a finite number.

② If both $\int_{-\infty}^a f(x)dx$ and $\int_a^{+\infty} f(x)dx$ are convergent, then we define

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{+\infty} f(x)dx.$$

Example 22

Evaluate the integrals

$$a) \int_{-\infty}^{+\infty} \frac{dx}{(x^2 + 1)^2} \qquad b) \int_{-\infty}^0 e^{2x}(x + 2)dx.$$

a) Set $x = \tan t$,

$$\begin{aligned} I &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 t dt = \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \cos 2t) dt \\ &= \frac{1}{2} \left(t + \frac{\sin 2t}{2} \right) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2}. \end{aligned}$$

Assume that $f(x)$ and $g(x)$ are **nonnegative** continuous functions on $[a, +\infty)$.

Theorem 3.1 (Comparison test 1)

If $f(x) \leq g(x)$ for $x \geq a$, then

- If $\int_a^{+\infty} g(x)dx$ converges, then $\int_a^{+\infty} f(x)dx$ also converges.
- If $\int_a^{+\infty} f(x)dx$ diverges, then $\int_a^{+\infty} g(x)dx$ also diverges.

Theorem 3.2 (Comparison test 2)

Assume that $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = k \in (0, \infty)$, then both $\int_a^{+\infty} f(x)dx$ and $\int_a^{+\infty} g(x)dx$ *either converge or diverge*.

We also write, $f(x) \sim kg(x)$ as $x \rightarrow +\infty$.

Proposition 3.1

- i) If $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 0$ and $\int_a^{+\infty} g(x)dx$ converges, then $\int_a^{+\infty} f(x)dx$ converges.
- ii) If $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = \infty$ and $\int_a^{+\infty} g(x)dx$ diverges, then $\int_a^{+\infty} f(x)dx$ diverges.

Example 23 (Test for convergence)

$$a) \int_{\pi}^{\infty} \frac{\sin^2 x}{x^2 + \ln x} dx$$

$$c) \int_1^{+\infty} \frac{dx}{\ln(1 + 3x)}$$

$$b) \int_2^{+\infty} \frac{(1+x)\sqrt{x}}{x^3 - 2x + 1} dx$$

$$d) \int_1^{+\infty} \left(e^{-\frac{4}{x^2}} - 1 \right) dx$$

Definition 6

If $\int_a^{+\infty} |f(x)|dx$ **converges**, then we say $\int_a^{+\infty} f(x)dx$ *converges absolutely*.

Note: $\int_a^{+\infty} f(x)dx$ *converges absolutely* $\Rightarrow$ *converges*.

Definition 7

If $\int_a^{+\infty} |f(x)|dx$ **diverges** and $\int_a^{+\infty} f(x)dx$ **converges**, then we say $\int_a^{+\infty} f(x)dx$ *converges conditionally*.

Example 24

Test for convergence

$$a) \int_2^{+\infty} \frac{x \cos 2x}{x^3 + 3} dx, \quad b) \int_{-\infty}^{+\infty} \frac{e^{-x^2} \cos x}{1 + x^2} dx.$$

$$c) \int_1^{+\infty} \frac{\sin x}{x} dx \quad d) \int_1^{\infty} \cos x^2 dx.$$

Improper integrals of type 2

Definition 8

- i) Let $f(x)$ be continuous on $[a, b)$ and is discontinuous at b . The *improper integral* of $f(x)$ on $[a, b)$:

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx.$$

if this limit exists (as a finite number). $x = b$ is called *singular point* of the integral.

- ii) Let $f(x)$ be continuous on $(a, b]$ and is discontinuous at a . The *improper integral* of $f(x)$ on $(a, b]$:

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx.$$

if this limit exists (as a finite number). $x = a$ is called *singular point* of the integral.

If the limit exists (as a finite number), we say the integral is *convergent*. Otherwise, we say it is *divergent*.

- iii) If f has a discontinuity at c , where $a < c < b$, and both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then

we define

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

Example 25

Evaluate the integrals

$$a) \int_0^2 \frac{dx}{\sqrt{4-x^2}}$$

$$b) \int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt[3]{\sin x}} dx$$

Remark 3.2

$$\int_a^b \frac{dx}{(b-x)^\alpha} \text{ converges} \Leftrightarrow \alpha < 1.$$

$$\int_a^b \frac{dx}{(x-a)^\alpha} \text{ converges} \Leftrightarrow \alpha < 1.$$

Assume $f(x)$ and $g(x)$ are **nonnegative** continuous functions on $[a, b)$ and $\int_a^b f(x)dx, \int_a^b g(x)dx$ have **unique singular point** $x = b$.

Theorem 3.3

If $0 \leq f(x) \leq g(x)$ for $x \in [a, b)$, then

- If $\int_a^b g(x)dx$ converges, then $\int_a^b f(x)dx$ also converges.*
- If $\int_a^b f(x)dx$ diverges, then $\int_a^b g(x)dx$ also diverges.*

Theorem 3.4

If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = k$, $0 < k < \infty$, then both $\int_a^b f(x)dx$ and $\int_a^b g(x)dx$ are either convergent or divergent.

Proposition 3.2

- If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = 0$ and $\int_a^b g(x)dx$ converges $\int_a^b f(x)dx$ converges.
- If $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = \infty$ and $\int_a^b g(x)dx$ diverges $\int_a^b f(x)dx$ diverges.

Example 26

Test for convergence

$$a) \int_0^1 \frac{\ln(1 + \sqrt{x})}{e^{\tan x} - 1} dx$$

$$b) \int_0^1 \frac{dx}{\sqrt[3]{x^2(1-x)}}$$

$$c) \int_0^{\infty} \frac{dx}{\sqrt{x + 2x^3}}$$

Definition 9

If $\int_a^b |f(x)| dx$ **converges**, then we say $\int_a^b f(x) dx$ *converges absolutely*.

Note: If $\int_a^b f(x) dx$ converges absolutely, then it also converges.

Definition 10

If $\int_a^b |f(x)| dx$ **diverges** and $\int_a^b f(x) dx$ **converges** then we say $\int_a^b f(x) dx$ *converges conditionally*.

Example 27

Test for convergence .

$$a) \int_0^{\infty} \frac{\sin 2x}{x\sqrt{x}} dx$$

$$b) \int_0^{\infty} \frac{\arctan x}{x\sqrt{x}} dx$$